



GS2962 vs. GS2972

Application Note

Purpose

Together with the GS2972 Data Sheet (GENDOC-047479), this document serves as a guide for using the GS2962.

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1. GS2962 Pin Out

1.1 Pin Assignment

	1	2	3	4	5	6	7	8	9	10
A	DIN 17	DIN 18	F/DE	H/HSYNC	CORE_VDD	PLL_VDD	LF	VBG	RSV	A_VDD
B	DIN 15	DIN 16	DIN 19	PCLK	CORE_GND	PLL_VDD	VCO_VDD	VCO_GND	A_GND	A_GND
C	DIN 13	DIN 14	DIN 12	V/VSYSN	CORE_GND	PLL_GND	PLL_GND	PLL_GND	CD_GND	SDO
D	DIN 11	DIN 10	STANDBY	SDO_EN/DIS	RSV	RSV	RSV	RSV	CD_GND	SDO
E	CORE_VDD	CORE_GND	RATE_SEL0	RATE_SEL1	CORE_GND	CORE_GND	TDI	TMS	CD_GND	CD_VDD
F	DIN 9	DIN 8	DETECT_TRS	CORE_GND	CORE_GND	CORE_GND	RSV	TDO	CD_GND	RSET
G	IO_VDD	IO_GND	TIM_861	20bit/10bit	DVB_ASI	SMPTE_BYPASS	IOPROC_EN/DIS	RESET	CORE_GND	CORE_VDD
H	DIN 7	DIN 6	ANC_BLANK	LOCKED	RSV	RSV	RSV	JTAG/HOST	IO_GND	IO_VDD
J	DIN 5	DIN 4	DIN 1	RSV	RSV	RSV	RSV	TCK	SDOUT_TDO	SCLK_TCLK
K	DIN 3	DIN 2	DIN 0	RSV	RSV	RSV	RSV	CORE_VDD	CS_TMS	SDIN_TDI

Figure 1-1: GS2962 Pin Assignment

1.2 Pin Descriptions

Table 1-1: GS2962 Pin Descriptions

Pin Number	Name	Timing	Type	Description
A1, A2, B1, B2, B3, C1, C2, C3, D1, D2, F1, F2, H1, H2, J1, J2, J3, K1, K2, K3	DIN17, DIN18, DIN15, DIN16, DIN19, DIN13, DIN14, DIN12, DIN11, DIN10, DIN9, DIN8, DIN7, DIN6, DIN5, DIN4, DIN1, DIN3, DIN2, DIN0	—	Input	PARALLEL DATA BUS Signal levels are LVCMOS / LVTTTL compatible.
A3	F/DE	Synchronous with PCLK	Input	PARALLEL DATA TIMING. Signal levels are LVCMOS / LVTTTL compatible. TIM_861 = LOW: Used to indicate the ODD / EVEN field of the video signal when DETECT_TRS is set LOW. The device will set the F bit in all outgoing TRS signals for the entire period that the F input signal is HIGH (IOPROC_EN/DIS must also be HIGH). The F signal should be set HIGH for the entire period of field 2 and should be set LOW for all lines in field 1 and for all lines in progressive scan systems. The F signal is ignored when DETECT_TRS = HIGH. TIM_861 = HIGH: The DE signal is used to indicate the active video period. DE is HIGH for active data and LOW for blanking. The DE signal is ignored when DETECT_TRS = HIGH.
A4	H/HSYNC	Synchronous with PCLK	Input	PARALLEL DATA TIMING. Signal levels are LVCMOS / LVTTTL compatible. TIM_861 is LOW: The H signal is used to indicate the portion of the video line containing active video data, when DETECT_TRS is set LOW. Active Line Blanking The H signal should be LOW for the active portion of the video line. The signal goes LOW at the first active pixel of the line, and then goes HIGH after the last active pixel of the line. The H signal should be set HIGH for the entire horizontal blanking period, including both EAV and SAV TRS words, and LOW otherwise. TRS Based Blanking (H_CONFIG = 1 _H) The H signal should be set HIGH for the entire horizontal blanking period as indicated by the H bit in the received TRS ID words, and LOW otherwise. TIM_861 = HIGH: The HSYNC signal indicates horizontal timing.

Table 1-1: GS2962 Pin Descriptions (Continued)

Pin Number	Name	Timing	Type	Description
A5, E1, G10, K8	CORE_VDD	—	Input Power	Power supply connection for digital core logic. Connect to 1.2V DC digital.
A6, B6	PLL_VDD	—	Input Power	Power supply pin for PLL. Connect to 1.2V DC analog.
A7	LF	—	Analog Output	Loop Filter component connection.
A8	VBG	—	Output	Bandgap voltage filter connection.
A10	A_VDD	—	Input Power	VDD for sensitive analog circuitry. Connect to 3.3VDC analog.
B4	PCLK	—	Input	PARALLEL DATA BUS CLOCK. Signal levels are LVCMOS / LVTTTL compatible.
B5, C5, E2, E5, E6, F4, F5, F6, G9	CORE_GND	—	Input Power	GND connection for digital logic. Connect to digital GND.
B7	VCO_VDD	—	Input Power	Power pin for VCO. Connect to 1.2V DC analog followed by an RC filter (see Figure 4-1 on page 13). VCO_VDD is nominally 0.7V.
B8	VCO_GND	—	Input Power	Ground connection for VCO. Connect to analog GND.
B9, B10	A_GND	—	Input Power	GND pins for sensitive analog circuitry. Connect to analog GND.
C4	V/VSNC	Synchronous with PCLK	Input	PARALLEL DATA TIMING. Signal levels are LVCMOS / LVTTTL compatible. TIM_861 = LOW: The V signal is used to indicate the portion of the video field/frame that is used for vertical blanking, when DETECT_TRS is set LOW. The V signal should be set HIGH for the entire vertical blanking period and should be set LOW for all lines outside of the vertical blanking interval. The V signal is ignored when DETECT_TRS = HIGH. TIM_861 = HIGH: The VSYNC signal indicates vertical timing. The VSYNC signal is ignored when DETECT_TRS = HIGH.
C6, C7, C8	PLL_GND	—	Input Power	Ground connection for PLL. Connect to analog GND.
C9, D9, E9, F9	CD_GND	—	Input Power	Ground connection for the serial digital cable driver. Connect to analog GND.
C10, D10	SDO, $\overline{\text{SDO}}$	—	Output	Serial Data Output Signal. Serial digital output signal operating at 2.97Gb/s, 2.97/1.001Gbs, 1.485Gb/s, 1.485 /1.001Gb/s or 270Mb/s.

Table 1-1: GS2962 Pin Descriptions (Continued)

Pin Number	Name	Timing	Type	Description												
D3	STANDBY	—	Input	CONTROL SIGNAL INPUT. Signal levels are LVCMOS / LVTTTL compatible. Puts the device in Standby mode to reduce power. HIGH to Standby.												
D4	SDO_EN/ $\overline{\text{DIS}}$	—	Input	CONTROL SIGNAL INPUT. Signal levels are LVCMOS / LVTTTL compatible. Used to enable or disable the serial digital output stage. When SDO_EN/ $\overline{\text{DIS}}$ is LOW, the serial digital output signals SDO and $\overline{\text{SDO}}$ are disabled and become high impedance. When SDO_EN/ $\overline{\text{DIS}}$ is HIGH, the serial digital output signals SDO and $\overline{\text{SDO}}$ are enabled.												
E3, E4	RATE_SELO, RATE_SEL1	—	Input	CONTROL SIGNAL INPUT. Signal levels are LVCMOS/LVTTTL compatible. Used to configure the operating data rate. <table><thead><tr><th>RATE_SELO</th><th>RATE_SEL1</th><th>Data Rate</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>1.485 or 1.485/1.001Gb/s</td></tr><tr><td>0</td><td>1</td><td>2.97 or 2.97/1.001Gb/s</td></tr><tr><td>1</td><td>X</td><td>270Mb/s</td></tr></tbody></table>	RATE_SELO	RATE_SEL1	Data Rate	0	0	1.485 or 1.485/1.001Gb/s	0	1	2.97 or 2.97/1.001Gb/s	1	X	270Mb/s
RATE_SELO	RATE_SEL1	Data Rate														
0	0	1.485 or 1.485/1.001Gb/s														
0	1	2.97 or 2.97/1.001Gb/s														
1	X	270Mb/s														
E7	TDI	—	Input	COMMUNICATION SIGNAL INPUT. Signal levels are LVCMOS/LVTTTL compatible. Dedicated JTAG pin. Test data in. This pin is used to shift JTAG test data into the device when the JTAG/ $\overline{\text{HOST}}$ pin is LOW.												
E8	TMS	—	Input	COMMUNICATION SIGNAL INPUT. Signal levels are LVCMOS/LVTTTL compatible. Dedicated JTAG pin. Test mode start. This pin is JTAG Test Mode Start, used to control the operation of the JTAG test when the JTAG/ $\overline{\text{HOST}}$ pin is LOW.												
E10	CD_VDD	—	Input Power	Power for the serial digital cable driver. Connect to 3.3V DC analog.												
F3	DETECT_TRS	—	Input	CONTROL SIGNAL INPUT. Signal levels are LVCMOS / LVTTTL compatible. Used to select external HVF timing mode or TRS extraction timing mode. When DETECT_TRS is LOW, the device extracts all internal timing from the supplied H:V:F or CEA-861 timing signals, dependent on the status of the TIM861 pin. When DETECT_TRS is HIGH, the device extracts all internal timing from TRS signals embedded in the supplied video stream.												

Table 1-1: GS2962 Pin Descriptions (Continued)

Pin Number	Name	Timing	Type	Description
F8	TDO	—	Output	COMMUNICATION SIGNAL OUTPUT. Signal levels are LVCMOS/LVTTL compatible. Dedicated JTAG pin. JTAG Test Data Output. This pin is used to shift results from the device when the JTAG/HOST pin is LOW.
F10	RSET	—	Input	An external 1% resistor connected to this input is used to set the SDO/ $\overline{\text{SDO}}$ output signal amplitude.
G1, H10	IO_VDD	—	Input Power	Power connection for digital I/O. Connect to 3.3V or 1.8V DC digital.
G2, H9	IO_GND	—	Input Power	Ground connection for digital I/O. Connect to digital GND.
G3	TIM_861	—	Input	CONTROL SIGNAL INPUT. Signal levels are LVCMOS / LVTTL compatible. Used to select external CEA-861 timing mode. When DETECT_TRS is LOW and TIM-861 is LOW, the device extracts all internal timing from the supplied H:V:F timing signals. When DETECT_TRS is LOW and TIM-861 is HIGH, the device extracts all internal timing from the supplied HSYNC, VSYNC, DE timing signals. When DETECT_TRS is HIGH, the device extracts all internal timing from TRS signals embedded in the supplied video stream.
G4	20bit/ $\overline{10\text{bit}}$	—	Input	CONTROL SIGNAL INPUT. Signal levels are LVCMOS/LVTTL compatible. Used to select the input bus width.
G5	DVB_ASI	—	Input	CONTROL SIGNAL INPUT Signal levels are LVCMOS/LVTTL compatible. Used to enable/disable the DVB-ASI data transmission. When set HIGH, the device will carry out DVB_ASI, word alignment, I/O processing and transmission. The $\overline{\text{SMPT}}\overline{\text{E_BYPASS}}$ pin must be set LOW. When $\overline{\text{SMPT}}\overline{\text{E_BYPASS}}$ and DVB_ASI are both set LOW, the device operates in data-through mode.

Table 1-1: GS2962 Pin Descriptions (Continued)

Pin Number	Name	Timing	Type	Description
G6	$\overline{\text{SMPTE_BYPASS}}$	—	Input	CONTROL SIGNAL INPUT. Signal levels are LVCMOS/LVTTL compatible. Used to enable / disable all forms of encoding / decoding, scrambling and EDH insertion. When set LOW, the device operates in data through mode (DVB_ASI = LOW), or in DVB-ASI mode (DVB_ASI = HIGH). No SMPTE scrambling takes place and none of the I/O processing features of the device are available when $\overline{\text{SMPTE_BYPASS}}$ is set LOW. When set HIGH, the device carries out SMPTE scrambling and I/O processing.
G7	$\text{IOPROC_EN}/\overline{\text{DIS}}$	—	Input	CONTROL SIGNAL INPUT. Signal levels are LVCMOS/LVTTL compatible. Used to enable or disable the I/O processing features. When $\text{IOPROC_EN}/\overline{\text{DIS}}$ is HIGH, the I/O processing features of the device are enabled. When $\text{IOPROC_EN}/\overline{\text{DIS}}$ is LOW, the I/O processing features of the device are disabled. Only applicable in SMPTE mode.
G8	$\overline{\text{RESET}}$	—	Input	CONTROL SIGNAL INPUT Signal levels are LVCMOS/LVTTL compatible. Used to reset the internal operating conditions to default settings and to reset the JTAG sequence. Normal mode (JTAG_EN = LOW). When LOW, all functional blocks will be set to default conditions and all input and output signals become high impedance. When HIGH, normal operation of the device resumes. JTAG test mode (JTAG_EN = HIGH). When LOW, all functional blocks will be set to default and the JTAG test sequence will be reset. When HIGH, normal operation of the JTAG test sequence resumes.
H3	$\overline{\text{ANC_BLANK}}$	—	Input	CONTROL SIGNAL INPUT. Signal levels are LVCMOS / LVTTL compatible. When $\overline{\text{ANC_BLANK}}$ is LOW, the Luma and Chroma input data is set to the appropriate blanking levels during the H and V blanking intervals. When $\overline{\text{ANC_BLANK}}$ is HIGH, the Luma and Chroma data pass through the device unaltered. Only applicable in SMPTE mode.
H4	LOCKED	—	Output	STATUS SIGNAL OUTPUT. Signal levels are LVCMOS / LVTTL compatible. PLL lock indication. HIGH indicates PLL is locked. LOW indicates PLL is not locked.

Table 1-1: GS2962 Pin Descriptions (Continued)

Pin Number	Name	Timing	Type	Description
H8	JTAG/ $\overline{\text{HOST}}$	—	Input	CONTROL SIGNAL INPUT. Signal levels are LVCMOS / LVTTTL compatible. Used to select JTAG test mode or Host Interface mode. When JTAG/$\overline{\text{HOST}}$ is HIGH, the Host Interface port is configured for JTAG test. When JTAG/$\overline{\text{HOST}}$ is LOW, normal operation of the Host Interface port resumes and the separate JTAG pins become the JTAG port.
J8	TCK	—	Input	COMMUNICATION SIGNAL INPUT. Signal levels are LVCMOS/LVTTTL compatible. JTAG Serial Data Clock Signal. This pin is the JTAG clock when the JTAG/$\overline{\text{HOST}}$ pin is LOW.
J9	SDOUT_TDO	—	Output	COMMUNICATION SIGNAL OUTPUT. Signal levels are LVCMOS / LVTTTL compatible. Serial Data Output / Test Data Output Host Mode (JTAG/$\overline{\text{HOST}}$ = LOW) This pin operates as the host interface serial output, used to read status and configuration information from the internal registers of the device. JTAG Test Mode (JTAG/$\overline{\text{HOST}}$ = HIGH) This pin is used to shift test results and operates as the JTAG test data output, TDO. Note: If the host interface is not being used leave this pin unconnected. IO_VDD = 3.3V Drive Strength = 12mA IO_VDD = 1.8V Drive Strength = 4mA
J10	SCLK_TCK	—	Input	COMMUNICATION SIGNAL INPUT. Signal levels are LVCMOS / LVTTTL compatible. Shared JTAG/HOST pin. Provided for pin compatibility with GS1582. Serial data clock signal. Host Mode (JTAG/$\overline{\text{HOST}}$ = LOW) SCLK_TCK operates as the host interface burst clock, SCLK. Command and data read/write words are clocked into the device synchronously with this clock. JTAG Test Mode (JTAG/$\overline{\text{HOST}}$ = HIGH) This pin is the TEST MODE START pin, used to control the operation of the JTAG test clock, TCK. Note: If the host interface is not being used, tie this pin HIGH.

Table 1-1: GS2962 Pin Descriptions (Continued)

Pin Number	Name	Timing	Type	Description
K9	$\overline{\text{CS_TMS}}$	—	Input	COMMUNICATION SIGNAL INPUT. Signal levels are LVCMOS / LVTTTL compatible. Chip select / test mode start. JTAG Test mode (JTAG/ $\overline{\text{HOST}} = \text{HIGH}$) $\overline{\text{CS_TMS}}$ operates as the JTAG test mode start, TMS, used to control the operation of the JTAG test, and is active HIGH. Host mode (JTAG/ $\overline{\text{HOST}} = \text{LOW}$), $\overline{\text{CS_TMS}}$ operates as the Host Interface Chip Select, $\overline{\text{CS}}$, and is active LOW.
K10	SDIN_TDI	—	Input	COMMUNICATION SIGNAL INPUT. Signal levels are LVCMOS / LVTTTL compatible. Shared JTAG/HOST pin. Provided for pin compatibility with GS1582. Serial data in/test data in. In JTAG mode, this pin is used to shift test data into the device. In Host Interface mode, this pin is used to write address and configuration data words into the device.
A9, D6, D7, D8, H5, H6, H7	RSV	—	—	These pins are reserved and should be left unconnected.
D5, F7	RSV	—	—	These pins are reserved, and should not be connected. If connection is necessary, connect to IO_GND.
J4, J5, J6, J7, K4, K5, K6, K7	RSV	—	—	These pins are reserved. Do not connect. If connection is necessary, connect to IO_VDD.

2. Electrical Characteristics

2.1 DC Electrical Characteristics

Table 2-1: GS2962 DC Electrical Characteristics

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Note
System							
+1.2V Supply Current	I _{1V2}	CORE_VDD	—	50	—	mA	—
		PLL_VDD	—	70	—	mA	—
		VCO_VDD	—	6	—	mA	—
+3.3V Supply Current	I _{3V3}	IO_VDD (3.3V)	—	10	—	mA	—
		A_VDD	—	3.2	—	mA	—
		CD_VDD	—	64	—	mA	—
+1.8V Supply Current	I _{1V8}	IO_VDD (1.8V)	—	5	—	mA	—
Total Device Power	P _D	STANDBY = 0	—	405	—	mW	—
		STANDBY = 1	10	125	—	mW	—
Digital I/O							
Input Logic LOW	V _{IL}	3.3V or 1.8V operation	—	—	0.3 x IO_VDD	V	—
Input Logic HIGH	V _{IH}	3.3V or 1.8V operation	0.7 x IO_VDD	—	—	V	—
Output Logic LOW	V _{OL}	I _{OL} = 5mA, 1.8V operation	—	—	0.2	V	—
		I _{OL} = 8mA, 3.3V operation	—	—	0.4	V	—
Output Logic HIGH	V _{OH}	I _{OL} = 5mA, 1.8V operation	1.4	—	—	V	—
		I _{OL} = 8mA, 3.3V operation	2.4	—	—	V	—
Serial Output							
Serial Output Common Mode Voltage	V _{CMOUT}	75Ω load, R _{SET} = 750Ω SD and HD mode	—	CD_VDD - ΔV _{SDO}	—	V	—

Notes:

1. This is 0.7V rather than 1.2V because there is a voltage drop across an external 105Ω resistor. See [Typical Application Circuit on page 13](#).
2. The 3.3V supplies must track the 3.3V supply of an external EQ and external CD.

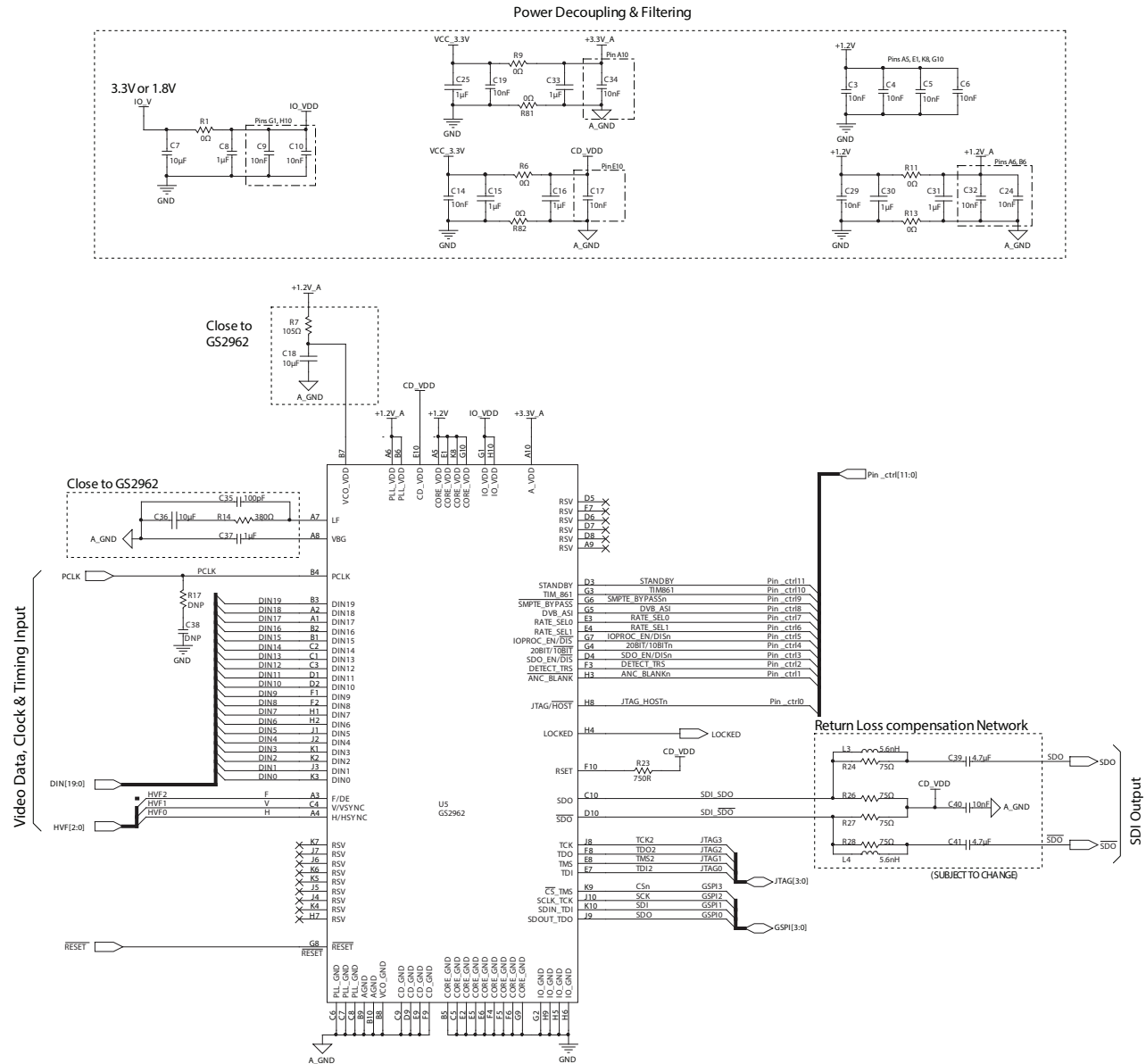
3. Audio References

The following is a detailed list of all of the sections in the GS2972 Data Sheet where Audio functions are mentioned. These sections, along with their associated figures and tables, do not apply to the GS2962 Transmitter:

- Key Features; 'Integrated audio Embedder for up to 8 channels of 48kHz audio'
- Description; 'In accordance with SMPTE 272M and SMPTE 299M, up to eight channels (two audio groups) of serial digital audio may be embedded into the video data stream. The input audio signal formats supported by the device include AES/EBU, I²S and serial audio. 16, 20 and 24-bit audio formats are supported at 48kHz synchronous for SD modes and 48kHz synchronous or asynchronous in HD, 3Gb/s modes.'
- 3.4 Serial Audio Data Inputs
- 3.5.2 Audio Embedding
- 3.5.3 SD Mode
 - 3.5.3.1 Operating Modes
 - 3.5.3.2 Audio Packet Detection
 - 3.5.3.3 Audio Packet Deletion
 - 3.5.3.4 Audio Mute (Default Off)
 - 3.5.3.5 Audio Channel Status
 - 3.5.3.6 Audio Crosspoint
 - 3.5.3.7 Audio Word Clock
 - 3.5.3.8 Channel & Group Activation
 - 3.5.3.9 Audio FIFO – SD
 - 3.5.3.10 Audio FIFO – HD and 3G
 - 3.5.3.11 Five-frame Sequence Detection – SD
 - 3.5.3.12 Frame Sequence Detection – HD
 - 3.5.3.13 Audio Control Packet Insertion – SD
 - 3.5.3.14 Audio Control Packet Insertion – HD and 3G
 - 3.5.3.15 Audio Data Packet Insertion
 - 3.5.3.16 Audio Interrupt Control
- Table 3-31: Video Core Configuration and Status Registers; Address 000_h Bit 13 (AUDIO_LEVELB_STREAM_2_1B) and Bit 10 (AUDIO_EMBED), Address 005_h Bits 10:0 (AUDIO_LATENCY), Address 02E_h Bit 5 (HOST_HD_AUDIO_RESET) and Bit 4 (HOST_SD_AUDIO_RESET) and Address 20E_h Bits 5:4 (AUDIO_INT_DS)
- 3.14 SD Audio Core
- 3.15 HD Audio Core Registers

4. Application Reference Design

4.1 Typical Application Circuit





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Contact Information

Semtech Corporation
200 Flynn Road, Camarillo, CA 93012
Phone: (805) 498-2111, Fax: (805) 498-3804
www.semtech.com